

Radiation, Magnitudes & Spectra

Problem Set

fizicyst

This problem set accompanies the *Radiation, Magnitudes & Spectra* handout. Section A has five objective (multiple-choice) questions; Section B has three subjective questions, pitched a little harder than Section A. Use $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$, $b = 2.898 \times 10^{-3} \text{ m K}$, $c = 3 \times 10^5 \text{ km/s}$, and $R_{\odot} = 6.96 \times 10^8 \text{ m}$ where needed, unless a question tells you otherwise.

Section A — Objective

O1. A star's temperature doubles while its radius stays fixed. By what factor does its luminosity change?

- (a) 2
- (b) 4
- (c) 8
- (d) 16

O2. A cool, thin gas cloud lies directly between us and a hot, continuously-emitting source. Looking through the cloud toward the source, we see:

- (a) a continuous spectrum
- (b) an emission spectrum
- (c) an absorption spectrum
- (d) no light at all, since the cloud blocks everything

O3. Two stars have a flux ratio $f_1/f_2 = 100$. Their magnitude difference $m_1 - m_2$ is:

- (a) -5
- (b) 5
- (c) -2.5
- (d) 100

O4. Which one of the following statements about the Hertzsprung–Russell diagram is **false**?

- (a) The main sequence is where stars spend the majority of their lives fusing hydrogen.
- (b) White dwarfs are hot but small, and therefore appear dim on the diagram.
- (c) Giants and supergiants are hotter than main-sequence stars of comparable luminosity.
- (d) A star's mass is the primary factor determining where it sits along the main sequence.

O5. Using the Rydberg formula, the hydrogen transition $n = 3 \rightarrow n = 2$ corresponds to a wavelength closest to:

- (a) 122 nm
- (b) 486 nm
- (c) 656 nm
- (d) 1875 nm

Section B — Subjective

S1. A main-sequence star has surface temperature $T = 9500$ K and radius $R = 2.5 R_{\odot}$. It is observed to have apparent magnitude $m = 6.2$. Take the Sun's absolute magnitude to be $M_{\odot} = 4.83$ and its effective temperature to be $T_{\odot} = 5775$ K.

- (i) Find the star's luminosity L , in units of $L_{\odot}$.
- (ii) Using $M - M_{\odot} = -2.5 \log_{10}(L/L_{\odot})$, find the star's absolute magnitude M .
- (iii) Using the distance modulus, find the distance to the star in parsecs.

S2. A main-sequence star's nuclear fuel supply is roughly proportional to its mass M , and its main-sequence lifetime is (fuel available)/(rate of fuel consumption), i.e. $\tau \propto M/L$.

- (i) Using the mass-luminosity relation $L/L_{\odot} = (M/M_{\odot})^{3.5}$, derive an expression for a star's main-sequence lifetime τ , relative to the Sun's lifetime $\tau_{\odot}$, in terms of $M/M_{\odot}$ alone.
- (ii) Taking $\tau_{\odot} \approx 10$ Gyr, estimate the main-sequence lifetime of a star with $M = 15 M_{\odot}$.
- (iii) Briefly explain, using your scaling from (i), why massive O and B stars are essentially never observed to be very old, even though the universe itself is far older than any individual O or B star's lifetime.

S3. A distant galaxy's $H\alpha$ line (rest wavelength $\lambda_0 = 656.3$ nm) is observed at $\lambda = 682.1$ nm.

- (i) Find the redshift z and the galaxy's recession velocity v (you may assume $v \ll c$).

- (ii) The galaxy's integrated starlight shows a blackbody-like continuum with an *observed* peak wavelength of 600 nm. Given that this light has been redshifted by the same factor z found in (i), find the *rest-frame* peak wavelength as emitted by the galaxy, and hence estimate the effective temperature of its stellar population using Wien's Law.
- (iii) Briefly explain why a cosmological redshift stretches a blackbody spectrum's wavelength scale without destroying its blackbody shape — i.e. the redshifted curve still looks exactly like a blackbody curve, just for a different (lower) temperature. What does this tell you about how λ_{max} must transform under redshift, and is that consistent with what you used in part (ii)?